

1. What is a model attribute?

It's the attribute(server side entry=key value pair : String, Object)

Purpose : to share the results of B.L from Handler --> View layer

Who creates --- Req handling Controller(handler) --prog supplied

Who sends it to whom : Handler --> D.S

After D.S gets actual view name from V.R :

D.S chks : are there any model attrs :

Yes : D.S saves model attrs under Request scope & then forwards the clnt to view layer .

NO : D.S forwards the clnt to view layer .

How to access these model attrs from JSP ?

`${requestScope.attrName}`

What are different ways for handler to add model attrs ?

1.1 Via o.s.w.s.ModelAndView?

o.s.w.s.ModelAndView :

It's a class representing a holder for model attrs + logical view name

Ctor :

`ModelAndView(String forwardViewName,String modelAttrName,Object modelAttrVal)`

eg : what will be the valid ret type of req handling method

String OR ModelAndView : ModelAndView

1.2 Any Simpler way to send model attributes from Handler(Controller) --> D.S ?

Use o.s.ui.Model : i/f ---

It represents holder (Map) of model attributes

How to add model attributes ?

`public Model addAttribute(String modelAttrName,Object modelAttrVal)`

eg : How to add multiple model attrs? : method chaining

Who will supply empty Model map (as the dependency)? : SC

IoC : DI

How to tell SC that handler method needs a model map ? : Simply add it as the arg of req handling method

When req handling controller rets string : logical view name (Handler implicitly rets all model attr map to D.S)

7. Handling request parameters in Handler ?

Request Handling method argument level annotation

`@RequestParam`

To bind request parameters to the method arguments of the request handling method

eg : `@RequestParam double price`

SC : `double price =Double.parseDouble(request.getParameter("price"))`

Reco : Match req param names with method arg names.

eg : URL : `http://localhost:8080/day13.1/test/product?nm=pen&qty=10&price=40.5&manuDate=2020-1-1`

Problem : HTTP 400 :

Bad request => some request data coming from client is invalid

Default date format : MM/dd/yyyy

To tell SC : use `@DateTimeFormat(pattern="yyyy-MM-dd")` annotation , along with `@RequestParam`

PRG pattern(Post-redirect-get pattern)

--- to avoid multiple submission issue in a web app.

Replace forward view(server pull) by redirect view (clnt pull) --a.k.a double submit guard.

How to replace default forward view by redirect view in spring MVC ?

Ans -- use redirect keyword.

eg : To redirect blogger to the home page -

```
return "redirect:/blogger/home";
```

Internals

D.S skips the V.R & sends temp redirect response to the clnt browser.

How ?

D.S invokes --- `response.sendRedirect(response.encodeRedirectURL("/customer/categories"));`

So clnt browser will send a next request ---with method=GET

URL --

`http://host:port/day13_boot/customer/categories`

URL after choosing a category :

`http://localhost:8080/day13_boot/customer/products;jsessionid=8210C7A572E0BF63BBB8176C40602A86?catId=1`

7. How to remember user details till logout?

Ans : add them as attributes in session scope.

How to access HttpSession in Spring?

Using D.I

How -- Simply add HttpSession as method argument of request handling method.

8. How to remember the details(attributes) till only the next request (typically required in PRG --redirect view)

Ans -- Add the attributes under flash scope.

(They will be visible till the next request from the same clnt)

How to add ?

Use i/f -- `o.s.w.s.mvc.support.RedirectAttributes` (sub i/f of Model i/f)

Method

```
public RedirectAttributes addFlashAttribute(String attrName, Object value)
```

How to access them in view layer in the next request?

via request scope attributes. `${requestScope.attrName}`

eg : In case of successful login --save user details under session scope(till user log out) & retain status msg only till the next request.

In case of invalid login --save status under request scope.

9. How to ensure all links(href)/form actions are created , relative to current context path + add URL rewriting support ?

1. Import spring supplied JSP tag lib. (OR use JSTL c:url)
(via taglib directive)
prefix ="spring"

2. Use the tag.

eg : <spring:url var="abc" value="/emp/list"/>

<form action="\${abc}" method="post">

...

</form>

NOTE : / => root of curnt web app.

What will be the URL , in absence of cookies ?

http://host:port/spring_boot/emp/list;jsessionid=tretqer5756765

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10. How to auto navigate the clnt to home page after logging out after some dly ?

Ans : By setting refresh header of HTTP response.

API of HttpServletResponse

public void setHeader(String name,String value)

name --- refresh

value --- 10;url=any url you want to redirect to (eg : home page url (root of web app))

10 : delay in seconds

How to get the root of curnt web app ?

API of HttpServletRequest

String getContextPath()